

Refocusing on the start and end of tourists' decision-making: Measuring thresholds and information loop limits

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ABSTRACT

In this era of abundant travel-related information, tourist decision-making processes have become increasingly complex and nonrational. This study focuses on the initiation and conclusion instances of decision-making between the pre-decision, decision-in-progress, and post-decision phases to capture the diversity of decisional behaviors. Building upon the Cyclic Model of Tourist Decision-Making, two new constructs, the Decision-Making Threshold (DMT) and Information Loop Limit (ILL), were validated with measurement tools developed, with which we empirically recognized the individual differences in the propensity to start and end decisions. Moreover, the dimensions of each construct were confirmed by incorporating the DMT for internal and external stimuli and the ILL based on cognitive capacity and willingness. The validation of the original theoretical concepts provides a basis for future research, including the examination of correlations among DMT, ILL, and other consumption-related behaviors. Four distinct decision-making segments are identified and practical implications are discussed.

1. Introduction

In an era of abundant online and offline travel-related information, potential tourists rapidly adjust their travel decision-making processes and do not always make rational choices (Wattanacharoensil & La-ornual, 2019). Some individuals make immediate travel decisions driven by inspiration (Dai, Wang, & Kirillova, 2022), whereas others, identified as perfectionists (Makgosa & Sangodoyin, 2018; Sproles & Kendall, 1986) or optimizing extenders (Mittal, 2017), meticulously gather, process, and evaluate as much information as possible before finalizing any purchase. Irrespective of decision-making mode, speed, and complexity, the travel decision-making process encompasses a series of sequential stages.

Schmoll (1977) proposed a widely accepted model of the travel decision-making process, consisting of four stages: travel desires, information search, assessment and comparison of travel alternatives, and decisions. Later models extend to include the travel experience and post-purchase evaluation stages (Mathieson & Wall, 1982; Moutinho, 1987). Subsequent research has explored the various phases of the decision-making process. For the *initiation* phase, recent years have witnessed diligent research focusing on the factors influencing tourists'

initial and repeat visit intentions (e.g., Gan, Shi, Filieri, & Leung, 2023; Huang, Li, & Lu, 2023; Maghrifani, Liu, & Sneddon, 2022; Peng, Yang, Fu, & Huan, 2023; Skard, Knudsen, Sjøstad, & Thorbjørnsen, 2021; Su, Lian, & Huang, 2020; Tassiello & Tillotson, 2020; Zhao, Shi, Huang, & Liu, 2023). Tourism scholars investigated the cognitive and affective aspects of information searching, processing, and evaluating alternatives *during* decision-making (e.g., Aliperti & Cruz, 2019; Castañeda, Rodríguez-Molina, Frías-Jamilena, & García-Retamero, 2020; Kang, Kim, & Park, 2021; Park & Jang, 2013; Townsend & Kahn, 2014). Regarding the *conclusion* phase, substantial efforts have been dedicated to investigating factors such as satisfaction, post-experience sharing, and memory retention (e.g., Agapito, Pinto, & Mendes, 2017; Bagheri, Guerreiro, Pinto, & Ghaderi, 2023; Kim, Ribeiro, & Li, 2022; Marques, Vinhas da Silva, & Antova, 2021). Bridging the gap between the end of one decision-making and the beginning of the next, studies have explored the influence of memories from previous experiences and intentions to revisit (Barnes, Mattsson, & Sørensen, 2016; Lim, Mohamed Jasim, & Das, 2024).

However, two pivotal turning points—the transition from pre-decision to decision-in-progress and the subsequent shift from decision-making to post-decision phases—have received relatively less

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academic attention. Extant models frequently employ broad descriptions like “travel needs/desires” and “making the decision” (e.g., Mathieson & Wall, 1982; Moutinho, 1987; Schmoll, 1977), which implies that the initiation and conclusion transition of a decision occur naturally. However, Fang (2023) posited a contrasting perspective, asserting that individuals consciously and unconsciously control the start and end of a decision. The start is based on whether the internal/external stimuli surpass a certain psychological threshold, termed the Decision-Making Threshold (DMT). Meanwhile, the determination to end a decision hinges on whether the internal/external constraints encapsulated in the Information Loop Limit (ILL) permit the ongoing processing of decision-related information. By leveraging conditional statements from computer science, the proposed cyclic model (Fang, 2023) accurately simulated both the initiation and conclusion stages of decision-making.

Focusing on the start and end turning points of decision-making, this study validated the DMT and ILL, developed reliable and valid measurement scales for the two constructs, and explored their attributes. This study is theoretically significant for two reasons. First, it completes the decision-making model by theorizing and validating new constructs to explain consumers’ tendencies to initiate and conclude decisions. This has drawn scholarly attention to two vital turning points in decision-making processes. Second, the new measurements for DMT and ILL could serve as valuable tools for further research, including the factors contributing to differences in DMT and ILL, correlations among DMT and ILL, consumers’ use of information and communication technologies, price sensitivity, purchase decisions, satisfaction, and brand loyalty.

For marketing practices, focusing on the start and end stages of the DMT and ILL constructs and their reliable measurement scales is also meaningful. They can provide insights into how individuals initiate decision-making and complete information processing, which can inform the development of effective marketing strategies and decision support tools. Furthermore, the distinction between the internal and external dimensions of the DMT and the willingness and cognitive capacity dimensions of the ILL can provide a more nuanced understanding of how decision-making begins and ends. This understanding can inform the timing and sequencing of marketing efforts, enabling marketers to strategically align their promotional activities or intervene with specific triggers that prompt individuals to initiate or finalize their decision-making.

2. Literature review

Tourist decision-making has undergone theoretical and empirical development for over half a century. Relaxing the assumption of perfect rationality is a key change in the development of tourist decision-making models. The traditional view of people as “straightforward, emotionless, algorithmic computer-like” decision-makers—akin to an idealistic rational choice model—is no longer widely accepted (Kahneman, 2003; Samson & Voyer, 2012; Stanovich, 2021; Stylos, 2022; Zajonc, 1980), and decision-making modeling has shifted to increasingly realistic and accurate models, such as bounded rationality (March & Simon, 1958; Ramos, Yamaka, Alorda, & Sriboonchitta, 2021; Simon, 1955).

The conceptualization of bounded rationality is influenced by the psychological processes of reasoning (Basel & Brühl, 2013). Neuroscientific experiments have empirically demonstrated the existence of various mental processes, of which two broad types are often observed: intuitive and analytical (Rizzolatti & Craighero, 2004; Verweij, Senior, Domínguez, & Turner, 2015). McCabe, Li, and Chen (2016) focused on two different processes that shape consumer behavior and reconceptualized a new dual-system tourist decision-making model that distinguishes heuristic, emotional, automatic, and intuitive System 1 from systematic, rational, deliberate, and complex reasoning System 2.

2.1. Cyclic model of decision-making

Fang (2023) adopted McCabe et al.’s (2016) conceptual model and proposed a cyclic model of tourist decision-making that considered Systems 1 and 2 as two ends of a continuum instead of two separate options. The varied mental processes along the continuum are distinguished by dimensions, including how decision-making starts, how alternative decision options are evaluated, and how information processing ends. The cyclic model is structured based on nested decision-making and information loops, delineating the iterative nature of the decision processes. The model adapts a conditional statement structure from computer science that manifests in two pivotal stages: the start of decision-making and the end of the information loop. These stages, which control the ingress and egress of the information loop, signify the start and end of the decision-making process, respectively.

The DMT is introduced at the start of the decision-making stage as “the minimum stimulus intensity that triggers the start of the decision-making” (Fang, 2023, p. 1171). The initiation of decision-related information processing depends upon the exceedance of the DMT, providing a quantifiable criterion for the transition into active decision-making. At the end of the information loop stage, the ILL concerns the “internal, external, and problem-related factors that stop decision-making entities from continuing their information processing” (Fang, 2023, p. 1173). Decision-making ends immediately with the ILL. In essence, the cyclic model emphasizes the need to explore the mechanisms dictating decision initiation and termination, which can elucidate a myriad of decision behaviors and clarify the formation of final decisions.

2.2. Start of decision-making (DMT)

Recent research has continued to focus on the initial stage of tourists’ decision-making processes. This phase is influenced by a range of factors, such as tourists’ perceptions of a destination’s social responsibility (Su, Lian, & Huang, 2020), subjective knowledge (Tassiello & Tillotson, 2020), and travel motivation (Maghrifani et al., 2022). Notably, Wang and Li (2022) conducted a meta-analytic review of 345 studies related to tourist consumption and discovered that external factors, such as information and quality, impact intention formation through internal mechanisms, such as trust and cognitive dissonance avoidance.

The role of technology in shaping visit intentions was further highlighted by Skard et al. (2021) and Ying et al. (2022), who examined the influence of virtual reality on travel intentions. Amidst the COVID-19 pandemic, scholars focus on factors such as risk, uncertainty, and ambiguity (Williams et al., 2022), travel shaming (Huang et al., 2023), risk message framing (Xie, Zhang, & Huang, 2023), and country stereotypes (Feng, Liu, & Li, 2022). In the realm of emerging trends in digital marketing, Gan et al. (2023) evaluated the impact of short video marketing on travel intentions, whereas Camilleri and Filieri (2023) investigated the role of online consumer reviews. Recent studies also focused on variables such as tourist happiness (Peng et al., 2023), tourism memory, and episodic future thinking (Zhao et al., 2023). These diverse studies collectively underscored the multifaceted nature of behavior in the early phase of tourist decision-making, highlighting the interplay of psychological, technological, and situational factors.

The earliest research on consumer decision-making process, traced back to John Dewey (1910), indicated the recognition of problem/s/needs as the starting moment of consumer decision-making, which was then widely accepted. Problem/need recognition is the crucial first stage in a consumer’s decision-making process (Bruner & Pomazal, 1988). Engel, Blackwell, and Miniard (1993, p. 508) defined problem recognition as the “... perception of a difference between the desired state of affairs and the actual situation sufficient to arouse and activate the decision process.” Based on this definition, Bruner (1985, 1986, 1987; Bruner & Pomazal, 1988) and Sirgy (1983, 1987) further examined how consumers initiate decision-making. Bruner (1985) classified

consumers into actual and desired state types by considering whether problem recognition was triggered by changes in an actual state or a desired state and demonstrated that the two types exhibit different consumption behaviors. Sirgy (1987) focused on four states when consumers compare current products with desired products and established that problem recognition is the strongest when negative incongruity was detected. In addition, an imbalance in homeostasis triggered problem recognition, and Bruner and Pomazal (1988) provided the following example:

After eating a meal, a person's need for food is satiated because the actual and desired states have been brought into balance. After an hour, as the food is digested, a slight difference may develop between the two states, although it is probably not large enough to be perceived. As several more hours pass, however, the size of the discrepancy increases beyond a "just noticeable difference" threshold and thus increases the probability of Problem Recognition. (p. 44).

For tourism consumption, the recognition of travel desire is commonly considered as the start of information processing. However, this assumption considers only the internal stimuli and not the external factors influencing the initiation of decision-making or the different forms of decision-making initiation under various circumstances. By integrating the stimulus-organism-response (S-O-R) and motivation-opportunity-ability (M-O-A) models, Laesser, Luo, and Beritelli (2019) proposed that individuals receive internal and external signals. Appetitive signals activate the evaluation process for one or more travel choice alternatives, whereas aversive signals neither activate nor inhibit evaluation. Evidently, the start of decision-making hinges not only on internal triggers but also on external signals, which may differ with individuals. Reflecting on our own behavior can illuminate differential responses to identical internal stimuli. For example, do we visit a travel agency immediately or browse travel agency websites as soon as we think about traveling? Some people may answer "yes" and others "no." Keshavarzian and Wu (2017) stated that "some participants revealed that they embarked on that trip solely because they found an air ticket to that destination on sale" (p. 552); however, this stimuli may not always surpass the DMT—individuals may respond differently to a homogenous external stimulus. Internal attention is the selective processing of internally generated signals, whereas external attention is the selection and modulation of incoming sensory information (Chun, Golomb, & Turk-Browne, 2011). Consequently, some individuals are more sensitive to spontaneously generated internal signals, whereas others primarily respond to external signals. Situational context further modulates this sensitivity by varying decision-making behaviors (Samson & Voyer, 2014). Distinctly categorizing these dimensions, not only deepens understanding of the heterogeneous psychological and cognitive processes at play among consumers but also lays the foundation for elucidating divergent decision-making styles. Moreover, by differentiating products, creating a salient internal signal, and crafting potent promotions or advertisements as external signals, marketers encourage individuals to consider products and services. Therefore, investigating how internal and external stimuli initiate decisions is both theoretically and practically meaningful.

However, Laesser et al.'s (2019) approach to distinguishing appetitive and inhibitory signals is a posteriori; this is because an identical signal affects different individuals differently. Therefore, instead of discussing whether an internal or external signal promotes or inhibits an individual's decision-making in each scenario, which may be expensive and of limited value, we propose studying how and why different individuals respond differently to the same stimuli. "Threshold"—mentioned in Bruner and Pomazal's (1988) example—captures such essential and long-overlooked differences among consumers. Our study adopted the DMT construct (Fang, 2023) in the tourist decision-making context and considered its dimensions corresponding to internal and external stimuli.

2.3. End of decision-making (ILL)

Every tourist's decision-making process ultimately enters its concluding phase, a critical stage characterized by a research focus on memory retention, satisfaction, and loyalty. Regarding the influential factors on tourist satisfaction and loyalty, Papadopoulou, Ribeiro, and Prayag (2023) examined the psychological determinants, emphasizing the impacts of perceived overcrowding and overtourism. Bagheri et al. (2023) focused on the effects of sense of well-being. Yang, Huang, and Huang (2021) analyzed the influence of sensory experiences on tourists' emotions and destination memories in addition to loyalty, underlining the emotional and experiential aspects of post-visit behaviors. Exploring the sharing of travel experiences, Su, Tang, and Nawijn (2021) investigated the effects of tourism activities, while Marques, da Silva, and Antova (2021) examined how the image of emerging destinations influences satisfaction and the intention to recommend and purchase products. Additionally, Agapito et al. (2017) explored the effects of tourists' sensory impressions on long-term memory and loyalty. Long-term remembered experiences significantly impact revisit intentions (Barnes et al., 2016). When recalling positive tourism memories, tourists had greater behavioral intentions, including revisit intention, word-of-mouth, and the desire to repeat the experience (Kim et al., 2022).

Research on the concluding phase of the decision-making process emphasizes the extended timeline of tourist consumer decision-making behaviors, which is vital for fostering long-term relationships and ensuring the sustainability of tourism businesses and destinations (Oppermann, 2000). Prior to the broadly defined conclusion phase in the aforementioned studies, a critical end moment focused more narrowly on information processing as the core of the decision-making process. Following the homo economicus assumption, this instance represents the point at which a decision-maker exhausts all available information to make an optimal decision. However, perfect rational decision-making is ideal, and actual decision-making typically ends when it is imperfect, considering the following realistic conditions:

1. Decision-makers do not pay attention to their own and others' proactive efforts to continue searching and are satisfied with selecting only existing feasible options.
2. Decision-makers lack relevant skills and, in some cases, make their own choices based on personal considerations.
3. Evaluating all options and selecting the best option require considerable time and money, which may not be worthwhile.

The total elapsed time from the beginning (needs recognition) to the end (purchase) of decision-making is defined as the total decision delay time (Greenleaf & Lehmann, 1995), and includes active decision time and time spent on all other activities. During active decision-making, decision-makers proactively collect and evaluate relevant information. Classical consumer decision-making theory states that once consumers gather and evaluate all necessary information, they make a final decision. However, few studies have examined how much information would be necessary or, in other words, how the total decision delay time would be brought to an end. With different lengths of total decision delay time, consumers exhibit different decision-making behaviors and make different final decisions under the influence of different factors.

Currently, it is widely acknowledged among researchers that our rationality is limited; thus, behavioral decision-making theories have gained popularity, with the Kahneman's Prospect Theory as one of the strongest perspectives for explaining irrational decision-making. Based on the dual-system theory (Kahneman, 2011), McCabe et al. (2016) proposed a dual-system framework, which then inspired Fang's (2023) cyclic model. Fang (2023) considered the two systems as two extremes: decision-makers carry out a certain degree of information processing until the ILL was reached, falling on a point on the continuum between the two extremes.

The existing literature delineates the pivotal role of cognitive capacity and willingness at the end of decision-making. In the era of abundant and complex information, individuals may reason that it is impossible to exhaustively assimilate all available information in decision-making. Impairments in cognitive capacity demarcate disparities in the extent to which available information is navigated, situating individuals along a spectrum of decision-making capacities (Peterson, Karlawish, & Largent, 2021). During the information processing stage, individuals navigate an escalating volume of information. Information overload occurs when the volume of available information transcends an individual's cognitive capacity (Bawden & Robinson, 2009; Hu & Krishen, 2019; Matthes, Karsay, Schmuck, & Stevic, 2020; Phillips-Wren & Adya, 2020). When confronted with equivalent volumes of information, individuals with "a lower level of cognitive capacity for dealing with relevant information tend to perceive higher levels of information overload" (Chen, Lin, & Cheng, 2022, p. 3) and adopt heuristics to save time and energy (Kahneman, 2011; McCabe et al., 2016), demonstrating a relatively lower ILL.

In terms of willingness to undertake exhaustive information processing in decision-making to determine the "best" choice, there is a salient distinction between satisficing and maximizing consumers.

Satisficers are content with a good enough choice, whereas maximizers strive for the optimal choice by conducting exhaustive searches for alternatives (Cheek & Schwartz, 2016; Schwartz et al., 2002). Limited willingness to continue information processing can be discerned from the impact of the level of involvement in decision behaviors. In high-involvement settings, individuals are prone to invest additional time and cognitive resources in the search and processing of travel-related information (Jun & Holland, 2012), demonstrating elevated ILL. Levels of ILL manifest heterogeneously among consumers in various decision-making situations. A deeper exploration of ILL can clarify the termination point of consumer decision-making, thereby enriching behavioral decision-making theory.

3. Overview of the research design

The main goal of the present study was to examine whether some individuals are more inclined to start decision-making processes, whereas some have a higher propensity to devote more time and effort to decision-making. Specifically, this study presents and discusses the construction, design, validation, and improvement of a measurement tool for individual differences in DMT and ILL. A series of studies were

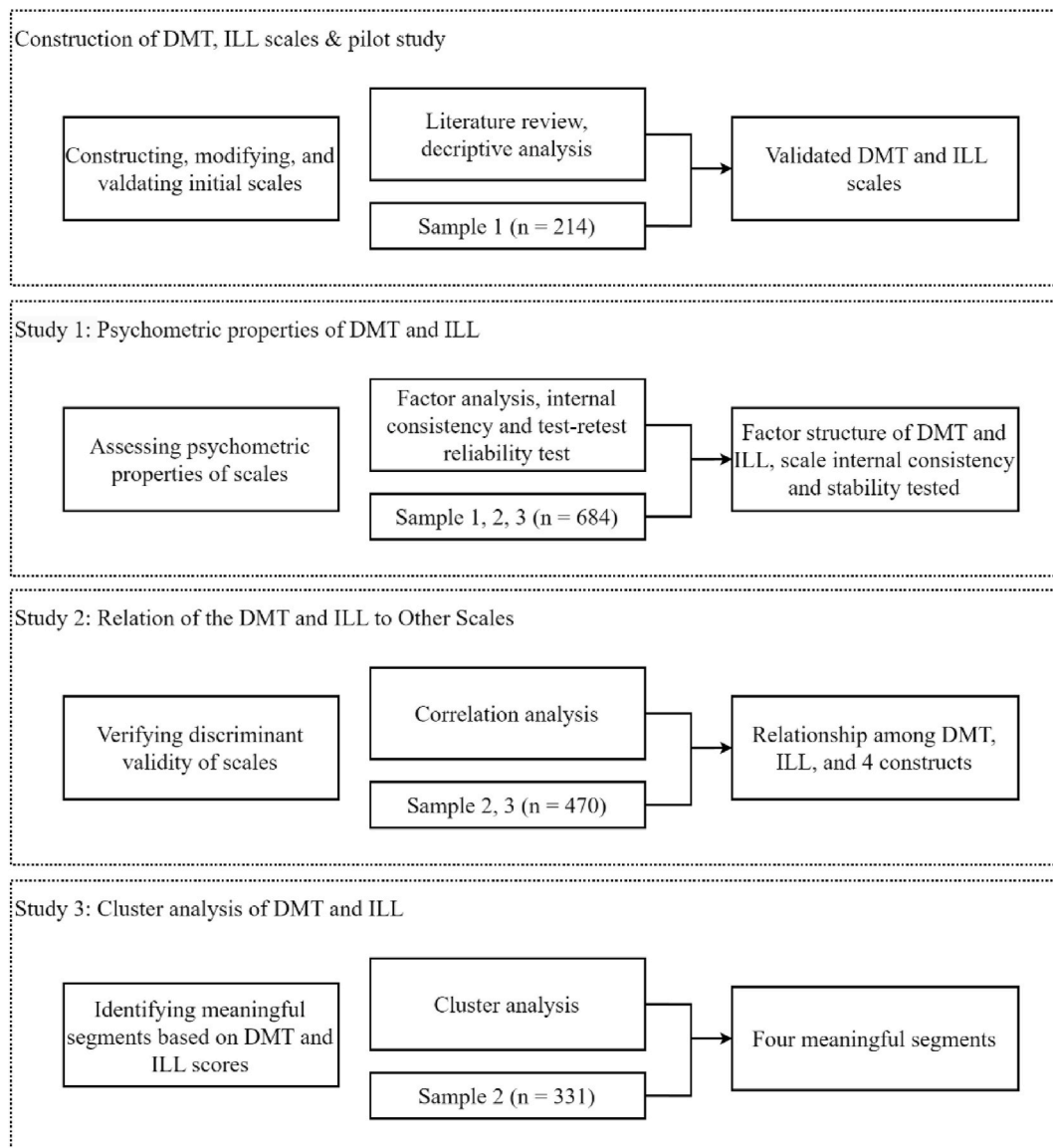


Fig. 1. Overview of the research design framework.

conducted on this topic. First, a pilot study modified and validated the initially designed measurement scales for DMT and ILL. Second, Study 1 demonstrated the factor structure, internal consistency, and stability of the two scales, assessing their psychometric properties. Specific techniques, such as exploratory and confirmatory factor analyses, internal consistency, and a test-retest reliability analysis, were utilized to show that the scales were sufficiently reliable and stable to capture individual differences. Third, Study 2 confirmed the discriminant validity of the two scales by examining the correlation between the DMT and ILL scores and other constructs related to consumer decision-making. Finally, Study 3 validated the practical utility of the measurement tools and highlighted their practical application through the identification of meaningful market segments with a cluster analysis based on DMT and ILL scores. The research framework is presented in Fig. 1.

A total of 684 valid questionnaires collected from three separate surveys were used in the pilot study and the three subsequent studies. For Sample 1, 214 valid questionnaires were gathered via wenjuan.com between July and August 2022, spanning across 28 of the 34 provincial-level administrative regions in China. Subsequently, to avoid potential bias stemming from the use of a single survey platform, Sample 2 utilized wenjuanxing.com and acquired 331 valid questionnaires between August and September 2022, covering 28 regions. Sample 3 comprised respondents from Sample 2, who were invited to participate in this further study two weeks after completing the initial questionnaire; 139 respondents across 26 regions consented and provided valid questionnaires.

wenjuan.com and wenjuanxing.com are expansive online survey platforms in China. They allowed to randomly distribute questionnaires nationwide via social media, email, and other channels. Respondents were compensated with monetary rewards upon completing questionnaires, distributed through the respective platforms.

4. Constructing the DMT and ILL scales

The core purpose of this phase of the study was to design and test measuring tools for the DMT and ILL. Items were initially generated based on Fang's (2023) definition, the theoretical discussion in the previous section, and existing measurement models for psychological constructs (e.g., Alós-Ferrer & Hügelschäfer, 2012; Jarvis & Petty, 1996). Subsequently, a pilot study was conducted to generate an initial scale to measure the two newly proposed constructs, the DMT and ILL. Through iterative testing and analysis, the initially generated measurement items were deleted, revised, or retained for further study.

4.1. Generation of the item pool

The DMT is the minimum intensity of a stimulus that triggers the start of decision-making. This is the critical point at which potential consumers receive a certain stimulus level, just enough to make them initiate decision-making behavior. "Stimulus" and "start of the decision-making" from the definition are two key points in drafting up each of the specific measurement items. For instance, for the item "I would not start planning future trips when recalling the pleasant feeling from previous traveling experiences" (回想起之前旅游经历带来的愉悦感并不会让我开始筹划将来的旅行), "Recalling the pleasant feeling from previous travel experiences" represents an internal stimulation, while "I would not start planning" demonstrates the scenario where the decision threshold is not reached, and the decision-making behavior is not initiated. Additionally, "I would not start travel consumption decision-making when reading information related to travel destinations" (看到旅游目的地的相关信息, 并不会让我开始旅游消费决策) describes the situation where the respondent receives external stimulation but the decision threshold is not exceeded. External and internal stimuli are distinguished, and we expect two dimensions in the construct.

The ILL is the upper limit of the information processing loop (including information recall, alternative evaluation, and information

collection) introduced by internal, external, and problem-related factors. The measurement tool concentrates on the subjective upper limit, and we expect that the construct will have two dimensions concerning the willingness and cognitive capacity of the decision-makers. The ILL items were designed based on definitions and related theories. The tourism context is provided in the items, for instance, "regarding tourism consumption decision-making ..." (关于旅游消费决策 ...) and "before making tourism decision ..." (在做出旅游决策之前 ...). The expression about willingness is "I am not willing to spend effort in tourism decision-making" (我不愿意花费精力进行旅游决策). As for cognitive capacity, items like "I cannot remember many pieces of information in tourism decision-making" (我无法在旅游消费决策过程中记住很多相关信息) were designed.

For empirical redundancy and considering that similar items can raise Cronbach's α without improving content validity (Clark & Watson, 2016), a few items with the same or highly similar meaning were carefully designed in different describing ways in the questionnaire. All items were measured on a 5-point Likert scale, labeled strongly disagree (1), disagree (2), not sure (3), agree (4), and strongly agree (5). For the DMT, a higher score indicated that the respondent had a lower tendency to initiate the decision-making process, indicating a higher DMT. For the ILL, a higher score indicated that the respondent tended to spend more time and effort in the decision-making process, indicating a higher ILL. To increase the readability and comprehensibility of the items and ensure clarity of meaning and high relevance to the constructs, two tourist behavior research experts were invited to examine each item statement thoroughly. Based on the consistent agreement, this study initially generated 15 items for the DMT and 17 for the ILL.

4.2. Pilot study: item screening and finalizing

To reduce the potential for acquiescence bias in responding, improve the content and face validity, screen out contradictory answers and cases where respondents did not answer seriously, reverse-scored items were included in the questionnaire. All positive/reverse-scored items appeared in random interspersed order. In total, 227 questionnaires were collected via the online survey platform wenjuan.com between July and August 2022. After screening and deleting all invalid responses, 214 valid responses (by 147 women and 67 men) were retained for further data analysis. For all items, responses ranged from 1 (minimum value) to 5 (maximum value). The scores of the negatively worded items were reversed to ensure consistency with the measured constructs. Thus, higher scores for all items indicate a higher DMT/ILL level for all respondents.

The DMT and ILL items were assessed following established qualitative and quantitative standards (e.g., Allen & Yen, 2001; Briggs & Cheek, 1986; Comrey, 1988; DeVellis & Thorpe, 2021). A consensus was reached between two experts and the authors who agreed that all items met the qualitative criteria, including (1) face validity, (2) clarity of meaning, and (3) a presumed unitary relationship with the proposed construct. Therefore, no items were removed based on the qualitative criteria.

The following guidelines were applied for the statistical criteria: (1) a mean greater than 2 and less than 4 on a 5-point scale, (2) a standard deviation of at least 1, (3) an item-total correlation greater than 0.3, and (4) an average inter-item correlation greater than 0.2. Upon evaluation, six DMT and eight ILL items were excluded because their standard deviations fell below 1, indicating insufficient heterogeneity. Three DMT items and one ILL item were removed based on Criteria (3) and (4), respectively.

None of the positively worded items in the DMT or ILL constructs met the statistical criteria. Specifically, positively worded items had lower standard deviations and significantly higher means than negatively worded items (after reverse scoring). This indicates a tendency for respondents to agree with positively worded items, regardless of their content, suggesting that the effect of acquiescence needs to be addressed

(Jackson, 1959; Welkenhuysen-Gybels, Billiet, & Cambré, 2003).

Furthermore, the pilot study analysis indicated that combining positively and negatively worded items did not improve scale reliability. This finding aligns with that of Vigil-Colet, Navarro-González, and Morales-Vives (2020), who suggested that scales that mix the two types of scoring may result in poorer psychometric properties. Suárez-Alvarez et al. (2018) also argued that when positively and negatively worded items were combined on the same scale, reliability was flawed and secondary sources of variance jeopardized the unidimensionality of the test. Consistent with these perspectives, the data from the pilot study revealed that the reliability of the scales containing both positively and negatively worded items (DMT: Cronbach's $\alpha = 0.729$; ILL: Cronbach's $\alpha = 0.873$) was lower compared to the scales with only negatively worded items (DMT: Cronbach's $\alpha = 0.815$; ILL: Cronbach's $\alpha = 0.932$).

Additionally, the exploratory factor analysis resulted in two main factors with eigenvalues greater than 1, each of which included either positively or negatively worded items. However, no meaningful factor structure emerged, suggesting that the inclusion of positively and negatively worded items led to correlated measurement errors (Alwin, 1973; Alwin & Krosnick, 1985; Andrews, 1984; Bentler, 1969; Costner, 1969; Green, 1988), causing the items to load on separate factors. Therefore, we removed all positively worded items from the DMT and ILL scales.

Finally, it was observed that removing item 15 from the ILL scale would lead to a slight improvement in Cronbach's α coefficient from 0.927 to 0.930. Considering the findings from the field of psychometrics that shorter scales are more effective (Credé, Harms, Niehorster, & Gaye-Valentine, 2012; Galesic & Bosnjak, 2009), item 15 was removed. This modification resulted in a concise and efficient questionnaire.

In summary, the questionnaire items underwent a rigorous screening process, resulting in the retention of seven items for the DMT scale and nine for the ILL scale (for the elimination process, please see Appendix A). Each scale included one positively worded item that identified invalid questionnaires rather than contributing to construct measurement. The DMT and ILL scales, consisting of the retained items, demonstrated good internal consistency, as indicated by Cronbach's α coefficient of 0.873 for the DMT and 0.930 for the ILL. These scales are now ready for thorough examination and validation in subsequent studies.

5. Study 1: psychometric properties of the DMT and ILL

The principal purposes of Study 1 were to (1) assess the factor structure of the DMT and ILL, (2) determine the internal consistency of the scales, and (3) assess the stability of participants' scores on the DMT and ILL. Following the pilot study, the first round of the formal survey was conducted, focusing specifically on the retained items. Using the online survey platform wenjuanxing.com, questionnaires were distributed randomly to participants from all over China. All participants provided permission to use the collected data for the research and were assured that their data would remain anonymous. This survey yielded 331 valid questionnaires (Sample 2: 54.08% women, 152; 45.92% men, 179). Additionally, 214 valid questionnaires from the pilot study (Sample 1) were included in the current study. Table 1 presents the retained items, their respective factor loadings, and Cronbach's α coefficients. A total of 543 valid responses were obtained with reliability and validity testing, which involved exploratory and confirmatory factor analysis, and internal consistency assessment. Two weeks after completing the initial questionnaire, participants from the first round of the formal survey were invited to complete the same survey again, yielding 139 valid questionnaires (Sample 3). This enabled us to analyze the test-retest reliability of the measurement tool and verify its stability over time.

Table 1

DMT and ILL scales, factor loadings, and Cronbach's α from two samples.

Item	Item wording	Factor loadings	
		Sample 1 (n = 214)	Sample 2 (n = 331)
DMT			
1	I do not initiate the process of making travel consumption decisions simply because I have the idea of traveling.	0.795	0.705
2	When I see travel-related advertisements, I do not start making travel consumption decisions.	0.752	0.705
3	Recalling the pleasure brought by previous travel experiences does not make me start planning future trips.	0.740	0.577
4	When I realize that I have a desire to travel, I do not initiate the process of making travel consumption decisions.	0.773	0.714
5	When I see information regarding travel destinations, I do not start making travel consumption decisions.	0.818	0.743
6	Even if friends discuss travel topics, I do not start making travel consumption decisions.	0.813	0.740
7	When I see travel-related information, I become aware of my travel needs and initiate the process of making travel consumption decisions. (R)	(Non-scoring item)	
	Cronbach's α	0.872	0.787
ILL			
1	I am unwilling to spend time and effort in information processing pertaining to travel consumption decisions. (R)	0.818	0.742
2	Making travel-related decisions does not require a great deal of information. (R)	0.741	0.633
3	I feel that spending too much time processing information pertaining to travel decisions is unnecessary. (R)	0.849	0.755
4	I do not have much time or energy to dedicate to information processing pertaining to travel consumption decisions. (R)	0.862	0.822
5	I do not let travel consumption decisions consume my time or energy. (R)	0.824	0.734
6	Travel consumption decisions involve too much information for me to consider. (R)	0.781	0.653
7	I only focus on a small amount of information during the travel consumption decision-making process. (R)	0.800	0.726
8	I cannot devote too much time and effort to information processing pertaining to travel consumption decisions. (R)	0.878	0.864
9	Before making travel consumption decisions, I spend a considerable amount of time and effort in information processing.	(Non-scoring item)	
	Cronbach's α	0.932	0.882

Note: Reversed scoring items are marked with (R).

5.1. Exploratory factor analysis

Based on the data analysis from Samples 1 and 2, no significant gender differences were found in either the DMT or ILL scores. An exploratory factor analysis was conducted to explore the underlying factors. The results of the Kaiser-Meyer-Olkin and Bartlett's tests indicated a significant level of correlation in the data, confirming its suitability for factor structure detection (Table 2).

The scree plot for DMT from Samples 1 and 2 indicates that the curves flattened after the second factor (see Appendix B). The first factor accounted for 61.2% of the variance in Sample 1 and 48.93% in Sample 2. The eigenvalue of the second strongest factor in both samples was <1 . However, when considering a two-factor solution, the explained variance increased to 71.35% and 64.24% in Samples 1 and 2, respectively. The DMT analysis suggests the presence of one dominant factor that explains a significant portion of the variance; however, more than 10%

Table 2

Kaiser-Meyer-Olkin test and Bartlett's Sphericity test results.

	KMO	Bartlett's Sphericity Test	
		Approx. Chi-Square	df
DMT Sample 1	0.883	555.172***	15
DMT Sample 2	0.814	506.669***	15
ILL Sample 1	0.906	1305.823***	28
ILL Sample 2	0.875	1308.022***	28

Note: ***, **, and * represent significance levels of 1%, 5%, and 10%, respectively.

of the variance is still accounted for by the second factor, indicating its potential importance.

Similarly, the scree plot for the ILL from Samples 1 and 2 exhibited flattened curves after the second factor, and the eigenvalues dropped below 1 when considering a two-factor solution. The first factor in Sample 1 accounted for 68.03% of the variance, and including a second factor increased the explained variance to 75.12%. In Sample 2, the single-factor solution explained 55.47% of the variance, whereas adding a second factor increased the cumulative explained variance by 11.10%. These results indicate that the ILL demonstrates a pattern comparable to that of the DMT, where a dominant factor explains a significant portion of the variance, but a second factor accounts for additional variance.

To further investigate the nature of the second factor in both the DMT and ILL constructs, we combined data from the two samples and conducted an exploratory factor analysis. Two factors were extracted for each construct, and the loadings of the DMT and ILL scale items for each factor were examined. The loadings were determined after applying promax rotation (Cureton & Mulaik, 1975; Hendrickson & White, 1964), and Table 3 presents the loadings for each item. We also examined other rotation methods, and the obtained loadings remained highly consistent across these methods. Hence, the results were not dependent on the rotation method employed.

Based on the literature review, our initial expectation was that both constructs would have two underlying dimensions. Specifically, we anticipated that there would be a DMT for internal and external stimuli and that decision-makers' willingness and cognitive capacity would comprise two dimensions of the ILL. Upon examining the factor loadings, we noticed that the allocation of items to different factors aligned well with the theoretical assumptions. In the DMT, the construct is bifurcated into two dimensions. Three items (DMT items 1, 3, and 4) were associated with factors related to internal stimuli, giving rise to the dimension named Decision-making Threshold for Internal Stimuli (DMTI). The remaining three items (DMT items 2, 5, 6) pertain to external stimuli, thereby forming the dimension named Decision-making Threshold for External Stimuli (DMTE).

By contrast, the ILL comprises two distinct dimensions. One dimension is correlated with the limitations induced by the decision-maker's willingness (ILL items 1, 2, 3, 5, and 7); hence, it is termed Information Loop Limit by Willingness (ILLW). The other dimension is connected to cognitive capacity (ILL items 4, 6, and 8), subsequently named the Information Loop Limit by Capacity (ILLC). We conducted a confirmatory factor analysis to assess factor structures of the DMT and ILL rigorously.

5.2. Confirmatory factor analysis

In this stage of the analysis, we compared alternative measurement models for the DMT and ILL (Fig. 2).

Models A and C assume that a single factor accounts for all variance in the scale items, whereas Models B and D suggest the existence of a second conceptual factor. The design of Models B and D was based on the findings of the exploratory factor analysis. By comparing the χ^2 difference between Models A and B, we found $\chi^2_{Model A} - \chi^2_{Model B} = 27.292$ (df = 1, $p < 0.0001$), indicating that the two-factor solution for the DMT fits the data better than the single-factor solution. Additional fit

Table 3

Loadings for two-factor solution for DMT and ILL items (Samples 1 and 2, n = 545).

Item	Item wording	Factor loadings	
DMT		Factor 1	Factor 2
1	I do not initiate the process of making travel consumption decisions simply because I have the idea of traveling.	0.568	
3	Recalling the pleasure brought by previous travel experiences does not make me start planning future trips.	0.883	
4	When I realize that I have a desire to travel, I do not initiate the process of making travel consumption decisions.	0.714	
2	When I see travel-related advertisements, I do not start making travel consumption decisions.		0.871
5	When I see information regarding travel destinations, I do not start making travel consumption decisions.		0.787
6	Even if friends discuss travel topics, I do not start making travel consumption decisions.		0.638
	Correlation between DMT Factor 1 and 2	0.901	
ILL		Factor 1	Factor 2
1	I am unwilling to spend time and effort in information processing pertaining to travel consumption decisions. (R)	0.684	
2	Making travel-related decisions does not require a great deal of information. (R)	0.864	
3	I feel that spending too much time processing information pertaining to travel decisions is unnecessary. (R)	0.728	
5	I do not let travel consumption decisions consume my time or energy. (R)	0.693	
7	I only focus on a small amount of information during the travel consumption decision-making process. (R)	0.721	
4	I do not have much time or energy to dedicate to information processing pertaining to travel consumption decisions. (R)		0.702
6	Travel consumption decisions involve too much information for me to consider. (R)		0.87
8	I cannot devote too much time and effort to information processing pertaining to travel consumption decisions. (R)		0.777
	Correlation between ILL Factor 1 and 2	0.834	

Note: The reverse-scored items are marked with (R). The scores were reversed for the reverse-scored items before factor analysis.

indices also supported this finding (refer to Table 4), with improvements in GFI, CFI, NFI, and NNFI and decreases in χ^2/df , RMSEA, and RMR when switching from Model B to Model A. Similarly, Model D was a better solution for the ILL at a significance level of 1%. The two-factor solution for the ILL demonstrated a better fit than the single-factor solution, as indicated by all the fit indices of Model D being superior to those of Model C.

5.3. Internal consistency

Table 5 presents the retained items for the DMT and ILL scales along with their means, standard deviations, and item-total correlations. It should be noted that both scales include a reversed item (marked with R) to identify invalid responders; however, the scores from these items were not used in the construct measurement. For the DMT scale, the item-total correlations ranged from 0.59 to 0.69, indicating that each of the six retained items demonstrated a reliable correlation with the overall DMT scale. Similarly, the item-total correlations for the ILL scale ranged from 0.67 to 0.84, demonstrating that all eight retained items were reliably correlated with the entire ILL scale.

Cronbach's α coefficients were calculated for both the DMT and ILL scales in Sample 1 (n = 214), Sample 2 (n = 331), and the overall collapsed dataset (n = 545). The results indicate that both scales

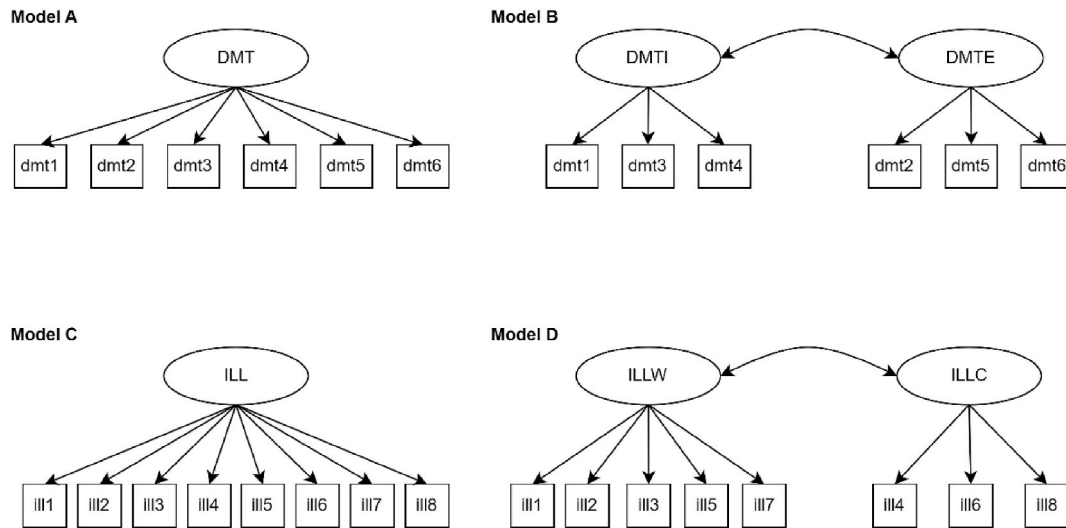


Fig. 2. Alternative DMT and ILL measurement models.

Table 4
Goodness of fit indexes for alternative confirmatory factor analysis models.

Model	χ^2/df	GFI	RMSEA	RMR	CFI	NFI	NNFI
	<3	>0.9	<0.10	<0.05	>0.9	>0.9	>0.9
A (DMT)	5.153	0.91	0.112	0.058	0.925	0.91	0.875
B (DMT)	2.385	0.963	0.065	0.036	0.978	0.963	0.958
C (ILL)	7.445	0.888	0.14	0.076	0.901	0.888	0.861
D (ILL)	2.185	0.969	0.06	0.039	0.983	0.969	0.974

The findings of the confirmatory factor analysis support the effectiveness of the two-factor solution for both the DMT and ILL, demonstrating a significant improvement in model fit.

demonstrated high internal consistency across all samples. For the DMT scale, Cronbach's α coefficients were 0.87, 0.79, and 0.86 for Sample 1, Sample 2, and the collapsed dataset, respectively. Similarly, for the ILL scale, Cronbach's α coefficients were 0.93, 0.88, and 0.92, respectively. See Tables 1 and 5 for more detailed information.

The mean inter-item correlation for the DMT scale was 0.51, while that for the ILL scale was 0.59. These values fall within the range of 0.20–0.60, indicating satisfactory homogeneity, breadth of coverage, and avoidance of excessive redundancy among the scale items (Briggs & Cheek, 1986).

5.4. Test-retest reliability

In the retest phase, approximately half the respondents from Sample 2 participated and completed the survey again after a two-week interval. In total, 139 valid questionnaires were collected, resulting in Sample 3. The test-retest reliability results demonstrated that the correlation coefficients between the scores from the two tests two weeks apart were 0.618 for the DMT scale and 0.659 for the ILL scale. These correlation coefficients were significant at the 1% level, which allowed us to reject the null hypothesis that the test results could not be replicated at different time points. The high correlation coefficients indicate that the DMT and ILL scales have high test-retest reliability. This suggests that the scales consistently measure the corresponding constructs over time, providing stable and reliable results.

6. Study 2: relation of the DMT and ILL to other scales

Building on the findings of Study 1, which established the reliability of the developed DMT and ILL scales in capturing individual differences, Study 2 aimed to examine the discriminant validity of these measures

further. The objective was to demonstrate that the DMT and ILL scales offer unique insights into consumer decision-making processes and are not redundant with existing constructs. Specifically, we selected four constructs closely connected to consumer decision-making behaviors. Constructs are the need to evaluate, faith in intuition, level of involvement, and mood.

To assess the discriminant validity of the DMT and ILL scales, we incorporated established measurement scales (see Table 6 for the sources of the scales) for the four constructs in the questionnaire administered to Samples 2 and 3. By analyzing the correlations between the DMT and ILL scores and existing constructs, we can evaluate the extent to which the DMT and ILL scales are unique. Additionally, we analyzed the relationships among the four constructs, the DMT, and the ILL to provide further insights into their interplay. Notably, these scales were initially in English and translated into Chinese by researchers proficient in both languages. The Consumer Involvement Profiles (CIP) scale was adapted to align with tourism.

6.1. Expected relationships

We anticipated that an individual's need to evaluate would have a negative correlation with the DMT and a positive correlation with the ILL. The need to evaluate is assumed to be one of the most pervasive and dominant human responses. This is measured using the NES, which refers to the degree to which an individual is motivated to evaluate and form judgments about various environmental stimuli (Jarvis & Petty, 1996, p. 172). The evaluation of alternatives is a crucial component of the decision-making process. We hypothesized that individuals who tend to assess the positive or negative qualities of an object/activity are more likely to respond to stimuli and initiate decision-making, leading to a lower DMT level.

Additionally, those with a higher need to evaluate are expected to engage in more elaborate and effortful cognitive processing. Kokkoris and Kühnen (2015) suggested that activating the need to evaluate may expand consumers' consideration sets. A high need to evaluate may also lead to the extension of other activities, such as evaluating the quality and relevance of information, weighing the advantages and disadvantages of various options, and considering the potential consequences of decisions. We expect that extended processing will result in a higher ILL.

Faith in intuition was expected to be negatively correlated with the ILL. In the decision-making process, individuals with a high level of faith in their intuition tend to rely more on intuitive thinking, which involves using their instincts and emotions to make judgments, rather than engaging in analytical thinking, such as weighing the pros and cons of

Table 5

Item means, standard deviations, and item-total correlations (Samples 1 and 2 collapsed, $n = 545$).

Item	Item wording	Mean	SD	ITC
DMT				
1	I do not initiate the process of making travel consumption decisions simply because I have the idea of traveling.	2.98	1.19	0.65
2	When I see travel-related advertisements, I do not start making travel consumption decisions.	2.83	1.12	0.64
3	Recalling the pleasure brought by previous travel experiences does not make me start planning future trips.	2.76	1.29	0.59
4	When I realize that I have a desire to travel, I do not initiate the process of making travel consumption decisions.	2.64	1.18	0.67
5	When I see information regarding travel destinations, I do not start making travel consumption decisions.	2.81	1.10	0.69
6	Even if friends discuss travel topics, I do not start making travel consumption decisions.	2.74	1.18	0.68
7	When I see travel-related information, I become aware of my travel needs and initiate the process of making travel consumption decisions. (R)		(Non-scoring item)	
Total scale		2.79	1.18	Cronbach's $\alpha = 0.86$
ILL				
1	I am unwilling to spend time and effort in information processing pertaining to travel consumption decisions. (R)	3.44	1.25	0.74
2	Making travel-related decisions does not require a great deal of information. (R)	3.38	1.23	0.68
3	I feel that spending too much time processing information pertaining to travel decisions is unnecessary. (R)	3.39	1.22	0.75
4	I do not have much time or energy to dedicate to information processing pertaining to travel consumption decisions. (R)	3.20	1.20	0.79
5	I do not let travel consumption decisions consume my time or energy. (R)	3.16	1.18	0.67
6	Travel consumption decisions involve too much information for me to consider. (R)	3.13	1.17	0.59
7	I only focus on a small amount of information during the travel consumption decision-making process. (R)	3.28	1.21	0.71
8	I cannot devote too much time and effort to information processing pertaining to travel consumption decisions. (R)	3.15	1.12	0.84
9	Before making travel consumption decisions, I spend a considerable amount of time and effort in information processing.		(Non-scoring item)	
Total scale		3.27	1.21	Cronbach's $\alpha = 0.92$

Note: The reverse-scored items are marked with (R). The scores were reversed for the reverse-scored items before factor analysis. ITC = item-total correlation.

decision alternatives. They would have lower incentives to collect and access information, leading to faster decision-making. The extreme case is making decisions without rational reasoning but solely based on intuition, which would result in a low ILL, especially in the willingness dimension.

Level of involvement is another construct that we expected to positively correlate with the ILL. This construct describes the extent of interest and concerns a consumer brings to a purchase decision task (Laurent & Kapferer, 1985). The more individuals are interested in or concerned with a specific purchase decision, the more likely they are to engage in extended problem solving, where they spend much time

Table 6

Sources of measurement models for constructs with potential relationship.

Construct	Measurement scale	Source
Need to evaluate	Need to Evaluate Scale (NES) Evaluative responding or the chronic tendency to engage in "the assessment of the positive or negative qualities of an object."	Jarvis and Petty (1996)
Faith in intuition	Faith in Intuition (FI) Reliance on intuitive decision-making.	Alós-Ferrer and Hügelschäfer (2012)
Level of involvement	Consumer Involvement Profiles (CIP) Extent of interest and concern that a consumer brings to bear on a purchase-decision task.	Laurent and Kapferer (1985)
Mood	Mood Short Form (MSF) State of emotional or affective arousal.	Peterson and Sauber (1983)

comparing different aspects, such as the features of the products, prices, and warranties, thus resulting in a higher ILL.

Research has demonstrated that positive affect promotes openness to information, the tendency to check hypotheses, and thoroughness when materials allow for that possibility and when individuals believe that putting in the effort will make a difference (Isen, 2008). Making decisions is perceived as a situation where more effort leads to better outcomes. This rationale suggests that a positive mood can increase an individual's motivation to engage in decision-making activities (negative correlation with the DMT) and consider a wider range of options (positive correlation with the ILL). However, other studies have suggested that positive mood (elicited by fast tempo music) can also lead to careless, superficial thinking and the use of heuristics (negative correlation with the ILL) rather than systematic cognitive processing (Palazzi, Wagner Fritzen, & Gauer, 2019). We expected mood to correlate with both the DMT and ILL and a negative correlation with the DMT, but existing studies have provided conflicting evidence regarding the relationship between mood and the ILL. Subsequently, we empirically investigated the direction of this correlation.

6.2. Results of study 2

Study 2 comprised 470 participants from Samples 2 ($n = 331$) and 3 ($n = 139$), with 257 female (54.68%) and 213 male (45.32%) participants. The four measurement scales included in the questionnaire demonstrated satisfactory levels of reliability, as indicated by Cronbach's α values exceeding 0.6. Notably, the NES, FI, and MSF scales had Cronbach's α values surpassing 0.7 (Table 7). Furthermore, correlations among the DMT and ILL constructs and the four aforementioned constructs were analyzed.

The results suggested that the relationship between the examined constructs was moderate, as no correlation exceeded 0.50. This finding indicates that the DMT and ILL constructs have good discriminant validity, meaning they measure different aspects of the decision-making process. The results also confirm the relationship between the various constructs and the DMT and ILL.

As expected, individuals with a higher need to evaluate tended to have lower DMT and higher ILL scores. Specifically, statistically significant negative correlations were observed between the need to evaluate and DMT for both the internal ($r = -0.264$, $p < 0.001$) and external dimensions ($r = -0.303$, $p < 0.001$), with a small to moderate effect size. A small positive correlation was also found between the need to evaluate and ILL, both for the willingness ($r = 0.226$, $p < 0.001$) and capacity dimensions ($r = 0.264$, $p < 0.001$).

Our expectations regarding the relationship between faith in intuition and ILL were confirmed by our results. The effect sizes for the relationships between the faith in intuition score and the ILL willingness ($r = -0.21$, $p < 0.001$) and capacity ($r = -0.142$, $p < 0.05$) dimensions were small. Willingness has a stronger relation with faith in intuition. This is because higher levels of faith in intuition indicate more trust in an

Table 7Cronbach's α of measurement scales and correlations among constructs and the DMT and ILL.

Scale	Cronbach's α	Correlation coefficient							
		DMTI		DMTE		ILLW		ILLC	
Need to Evaluate (NES)	0.727	−0.264	***	−0.303	***	0.226	***	0.264	***
Faith in Intuition (FI)	0.841	0.130	**			−0.210	***	−0.142	***
Consumer Involvement Profiles (CIP)	0.604					0.217	***		
Mood Short Form (MSF)	0.783	−0.333	***	−0.384	***	0.398	***	0.429	***

Note: ***, **, and * represent significance levels of 1%, 5%, and 10%, respectively.

individual's intuition, leading to the belief that decisions made based on intuition, even with less effort and time, will be correct, thereby reducing the motivation to continue information processing. Interestingly, the results demonstrated a small positive correlation between faith in intuition and the internal dimension of the DMT ($r = 0.13$, $p < 0.05$), which was unexpected. This connection requires further testing, and additional research is required to better understand the mechanism underlying this correlation.

The level of involvement was positively correlated with ILL in the willingness dimension ($r = 0.217$, $p < 0.001$). The more involved an individual is with a particular decision, the more motivated they are to make a satisfying or less regrettable decision, leading to a higher willingness to access and process information. No significant correlation was found between the level of involvement and the ILL in the capacity dimension or between the level of involvement and the DMT.

The expected negative relationship between mood and DMT was supported by our findings (both internal $r = -0.333$, $p < 0.001$ and external $r = -0.384$, $p < 0.001$ dimensions demonstrate a moderate negative correlation with mood). However, our results demonstrate a positive and moderate relationship between mood and ILL (willingness $r = 0.398$, $p < 0.001$; capacity $r = 0.429$, $p < 0.001$), indicating that the positive influence of mood dominates tourist decision-making.

Our findings highlight that an individual's tendency to initiate decision-making and engage in thorough information processing is connected to various factors. Multiple factors likely contribute to the prediction of the DMT and ILL. Further research is needed to explore the full range of factors that affect the initiation and conclusion of decision-making in tourism and to understand the complex interplay between them.

7. Study 3: cluster analysis of the DMT and ILL

Study 3 was designed to validate the practical utility of the measurement tools, thereby bridging the theoretical constructs with their practical application in the realm of consumer decision-making. This phase of the research, building upon the established reliability and validity of the DMT and ILL measures, was specifically focused on clustering participants based on their DMT and ILL scores. This clustering helps to identify distinct consumer segments, which would offer essential insights for marketers and decision-makers, enabling them to comprehend and address the diverse decision-making styles of consumers effectively.

7.1. Data analysis

A cluster analysis was conducted on data from Sample 2, which included 331 participants. The participants were clustered to create homogeneity within clusters and heterogeneity between clusters. A hierarchical cluster analysis was used to determine the number of clusters. The K-means cluster method, which is commonly used in tourist segmentation research (e.g., Cha, McCleary, & Uysal, 1995; Formica & Uysal, 1998; Keng & Cheng, 1999; Madrigal & Kahle, 1994; May, Bastian, Taylor, & Whipple, 2001), was used to classify samples according to the DMT and ILL.

Statistical tests were employed to profile clusters based on

demographic and psychographic characteristics, travel experiences, and tourism-related decision-making experiences. Relevant information was collected using the questions included in the questionnaire. Different statistical tests were employed based on the nature of the variables (metric or nonmetric). A chi-squared analysis was used to explore the differences between clusters regarding categorical variables, including demographics, travel experiences, and tourism-related decision-making experiences. The Kruskal-Wallis rank sum test was employed to assess potential differences among the clusters in terms of psychographic characteristics, which are metric variables.

A hierarchical cluster analysis was implemented using Euclidean distance measures based on the DMTI, DMTE, ILLW, and ILLC scores. The analysis revealed four distinct segments within the respondent pool. The distributions of the scores across these clusters are illustrated in Fig. 3.

The ANOVA results revealed that all four variables contributed to the differentiation of the four clusters ($p < 0.001$). Subsequently, Sidak *post hoc* tests were used to examine the differences between the clusters with respect to each variable. Statistically significant differences were found between each pair of clusters at the 0.001 level for all variables, except for the DMT external dimension, where the difference between Clusters 1 and 2 was significant at 0.05 level ($p = 0.046$), and no significant difference was found between Clusters 3 and 4.

The chi-square analysis and the Kruskal-Wallis rank sum test revealed significant differences among the clusters (Table 8). Four meaningful segments were identified and labeled: (1) rational frequent travelers, (2) travel career ladder climbers, (3) less-experienced low-information decision-makers, and (4) senior cautious decision-makers.

7.2. Illustrations of the clusters

Cluster 1 (rational frequent travelers) was characterized by a low DMT and high ILL. Among the four clusters, they were the most experienced tourists with rich travel-related decision-making experiences. Many married participants with underage children belonged to this cluster. The rational frequent travelers had high evaluation and cognitive needs, were in a good emotional state, and had relatively low levels of intuitive trust.

Cluster 2 (travel career ladder climbers) was characterized by a low DMT, moderate information processing capacity, moderate travel experience, travel-related decision-making experience, and a concentration of decision-makers in the 18–30 age group. Compared with Cluster 1, their cognitive and evaluation needs were lower. With increasing age and travel experience, individuals in Cluster 2 may gradually shift toward Cluster 1.

Cluster 3 (less-experienced, low-information decision-makers) was characterized by a moderate DMT and low ILL. They had relatively few travel-related decision-making experiences. Many single and childless individuals were concentrated in this cluster, which is reasonable because they faced fewer constraints. Unlike decision-makers with children, they do not have to consider the arrangement of children on the trip, resulting in a lower need for information. Cognitive and evaluation needs were even lower than those in Cluster 2. They tended to be in a poor emotional state, did not easily initiate decisions, and were prone to impulsive heuristic decision-making.

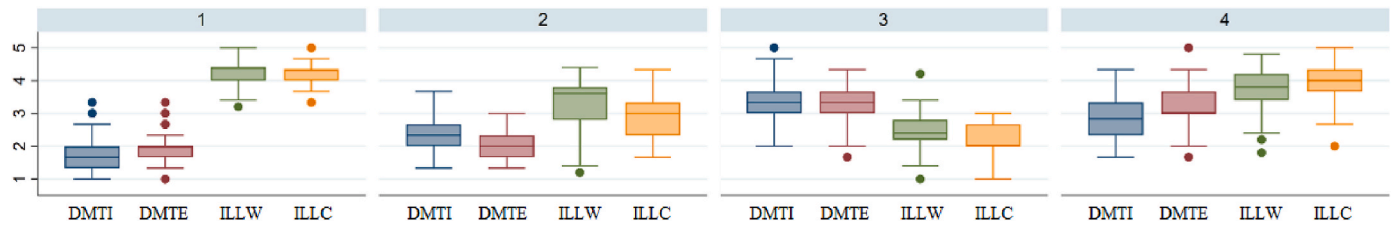


Fig. 3. Box-plot for DMT and ILL scores in four clusters.

Table 8

Pairwise differences among clusters.

Constructs	Chi-square/RankMeans Difference		
	Cluster 1	Cluster 2	Cluster 3
Cluster 2			
Travel decision-making frequency	Chi2(3) = 8.47	*	
Need to evaluate	RMD = 52.04	***	
Need for cognition	RMD = 88.58	***	
Faith in intuition	RMD = 39.61	**	
Mood	RMD = 62.97	***	
Cluster 3			
Family Life Cycle Stage	Chi2(4) = 11.38	*	
Travel frequency	Chi2(3) = 8.73	*	
Travel decision-making frequency	Chi2(3) = 21.68	***	
Need to evaluate	RMD = 75.54	***	RMD = 23.49
Need for cognition	RMD = 140.62	***	RMD = 52.03
Faith in intuition	RMD = 37.39	**	RMD = 2.22
Level of involvement	RMD = 43.37	**	RMD = 40.08
Mood	RMD = 119.12	***	RMD = 56.14
Cluster 4			
Age group	Chi2(3) = 9.89	*	
Need to evaluate	RMD = 47.7	***	RMD = 27.83
Need for cognition	RMD = 77.76	***	RMD = 62.86
Level of involvement			RMD = 28.42
Mood	RMD = 47.66	***	RMD = 71.46

Notes: ***, **, and * represent significance levels of 1%, 5%, and 10%, respectively. Chi² = Chi-square. RMD = rank-mean difference.

Cluster 4 (cautious senior decision-makers) was characterized by a moderate DMT and high ILL. In a comparison of sociodemographic characteristics, Cluster 4 had a relatively older age distribution. Compared to Cluster 3, their cognitive and evaluation needs were higher but lower than those of Cluster 1. Clusters 4 and 2 were similar to some degree, but when comparing external and internal DMT as well as ILL willingness and capacity, Cluster 4 had a relatively higher external threshold and was less likely to initiate decisions due to external stimuli, while Cluster 2 was less likely to initiate decisions due to internal stimuli. Cluster 4 had an equal level of information processing willingness and capacity, while Cluster 2 had lower information processing capacity than willingness, indicating that for Cluster 2, information processing capacity may be a constraining factor preventing ongoing information processing.

7.3. Summary

Study 3 provided compelling evidence for the practical utility of the DMT and ILL measurement scales. The clusters identified based on DMT and ILL scores exhibited distinct profiles regarding demographic and psychographic characteristics, travel experiences, and tourism-related decision-making experiences. A cluster analysis identified four meaningful segments: rational frequent travelers, travel career ladder climbers, less-experienced low-information decision-makers, and cautious senior decision-makers.

By understanding the characteristics of each segment, practitioners can develop targeted decision-making intervention strategies to provide appropriate guidance and support to enhance the decision-making experience of their target audience.

Overall, the findings of Study 3 highlight the practical applicability of the DMT and ILL measurement scales and demonstrate their potential to guide marketing and decision intervention efforts in the tourism industry.

8. General discussion and conclusions

8.1. Discussions and implications

Existing studies characterize the start of decision-making as a response to “travel needs/desires” (Mathieson & Wall, 1982; Moutinho, 1987; Schmoll, 1977) and presume that decision-making ends naturally. However, we consider the initiation of decision-making as a response to external or internal stimuli. Drawing on established literature, we inferred that the DMT encapsulates two distinct dimensions corresponding to internal (DMTI) and external signals (DMTE) and empirically validated this premise through a factor analysis. We found that the termination of decision-making was constrained by the decision-maker’s cognitive capacity (ILLC) and willingness (ILLW).

Considering the DMTE dimension, the mechanism by which external signals stimulate the initiation of decision-making requires further exploration. While extant research has focused on the relationship between external stimuli such as social information cues (Cheung, Xiao, & Liu, 2014), embodied experience (Wen & Leung, 2021), marketing communication (Song, Ruan, & Jeon, 2021), and purchasing decisions, a more granular exploration of the interaction between external signals and DMTE at the initiation phase of the decision process is pivotal. Once signals surpass an individual’s DMTE and initiate information processing, the product or service, though not yet purchased, enters the individual’s awareness set, significantly enhancing the possibility of future purchases. Studies on the effect of time of day on purchasing (e.g., Collinson, Mathmann, & Chylinski, 2020; Gullo, Berger, Etkin, & Bollinger, 2019; Yang, Wang, Li, Chen, & Yu, 2022) imply that consumers’ DMTI fluctuates at different times of the day. A profound understanding of how the DMTI differs across different environments and time periods has substantial theoretical and practical implications. The two ILL dimensions highlight that decision interventions (e.g., those that decrease willingness or compel the individual to reach cognitive capacity) may have different impacts on individuals. It is worth exploring what influences the differences in decision-making behaviors and final choices when the ILLW or ILLC dimensions lead to the end of a decision.

Theoretically, this study contributes to the field of consumer behavior by focusing on often overlooked instantaneous moments at the start and end of decision-making. These moments are less scrutinized than the more extensively researched pre-decision, decision-in-progress, and post-decision phases. By emphasizing the turning points among these three traditionally studied stages, this study validates two novel and distinct constructs: DMT and ILL. The empirical evidence demonstrated varied levels of DMT and ILL among decision-makers, supporting the existence of individual differences in the propensity to start and end decision-making (see Fig. 4). This adds credibility to the newly introduced cyclic model of tourist decision-making (Fang, 2023), which emphasizes that decision-makers intentionally or unintentionally control the start and end of the decision-making process. The constructs enable a more effective interpretation of the spectrum of behaviors observed in this domain, from immediate decisions (Dai et al., 2022) to those characterized by perfectionism (Makgosa & Sangodoyin, 2018; Mittal, 2017; Sproles & Kendall, 1986).

This study introduces new measurement tools that can significantly enhance the scope of research on consumer decisions. The use of these tools amplified the explanatory and predictive power of the cyclic decision-making model. Reliable measurement tools for the two constructs offer opportunities for further research, including quantitative analysis of the triggers and influencing factors in consumer decision-making and comparative studies of decision processes across different individuals or scenarios, thereby effectively bridging the gap between theoretical research and practical applications.

Marketing practices have ventured into methods for stimulating consumption, such as incorporating strategies related to the design of layouts, fonts, letter sizes, colors, and visuals to enhance consumer engagement. The focus of this study on the start and end instances of the decision-making process presents new points of consideration for the marketing professionals when formulating strategies. For instance, at the beginning of a decision, the goal of marketing efforts is to breach the DMT and guide the individual into the information processing phase. Stimuli designed to break through DMT for various demographics can be developed. Exploring measures to lower audience DMT is also vital. Regarding the decision's conclusion, marketers sometimes aim to hasten the decision-making process (e.g., when a consumer shows a preference or intention to purchase), while at other times, they seek to extend the information processing duration of the decision-maker (e.g., when a consumer's purchase intention is weak but could be strengthened by further understanding the product). In these scenarios, marketers must apply differentiated interventions based on their assessment of the buyer's intent. For the former, time-limited discounts may be effective, whereas recommending similar products to encourage continued comparison and deliberation may be more appropriate for the latter.

For commercial purposes, tracking consumer behavior, commonly used by online shopping platforms, can record the frequency with which consumers click on push ads, the elapsed time between product click-through and purchase, and their comparison of similar products. By including the DMT and ILL characteristics, these trackers facilitate the development of robust user profiles, which subsequently inform tailored marketing strategies to accurately influence potential consumers. Interventions can be formulated to ignite/avoid information overload, dissuade/enhance a decision-maker's willingness to process further information, and guide decision-makers to conclude the information loop at opportune moments, resulting in decisions that are favorable to marketers. In instances where consumers opt to not proceed with a purchase (e.g., abandoning a product detail page during online shopping), the immediate introduction of new information (e.g., subsidies or discounts) elevates the stimuli above the DMT, rekindling consumer engagement in the purchasing decision-making process.

Furthermore, marketing practitioners can develop and implement targeted marketing strategies based on their understanding of the distribution of individuals with various decision-making initiation and conclusion tendencies across populations. Four distinct segments of

decision-makers were identified: rational frequent travelers, travel career ladder climbers, less-experienced low-information decision-makers, and cautious senior decision-makers. For the third segment—the less-experienced low-information decision-makers—more effort should be made to stimulate travel desires and present comprehensive information in a swift and intuitive manner to compel impulsive decisions. However, these efforts may not substantially impact the “rational frequent travelers” segment. This group places a high value on the availability of detailed information and thorough processing. Emails enriched with hyperlinks related to a product or service may be a more suitable approach.

8.2. Limitations and further studies

One potential criticism of the newly constructed DMT and ILL scales is their reliance on self-reported data, which is a common limitation of psychographic scales. The research process was rigorous and comprehensive, and efforts were made to ensure the stability and reliability of the measurement scales; however, acknowledging the limitations associated with self-report measures is crucial. While negative wording was used for items to eliminate the effect of acquiescence, there remains the possibility of response biases, such as social desirability, which cannot be entirely eliminated in self-report data. To address this limitation, future studies should consider incorporating additional data sources, such as objective measures or behavioral observations, to complement self-reported findings and enhance the validity and robustness of the results.

Another potential limitation is the study's exclusive focus on respondents from China, which may restrict the direct generalizability of the findings to other cultural contexts. Future studies should broaden their participant base to include individuals from diverse cultural backgrounds. By doing so, the cross-cultural applicability and robustness of the DMT and ILL scales can be examined, and potential variations in decision-making tendencies across cultures can be explored further.

In scenarios involving long-haul versus short-haul travel and first-time versus repeat travel planning, tourist decision-making behaviors frequently exhibit variations. It is imperative to recognize the potential impacts of these variances on the start and end of tourists' decision-making processes. Investigating decision initiation and conclusions within diverse tourism-planning contexts has emerged as a compelling research avenue for further exploration.

Finally, although the current research utilized the DMT and ILL scores and discovered meaningful segments, future research may need to consider the predictive validity of newly constructed measures by examining their ability to predict relevant outcomes. This approach provides additional evidence for the validity of the measures and increases confidence in their usefulness in future research. Future research may also examine the antecedents and consequences of the tendency to start or end decision-making, which can clarify the mechanisms underlying different decision-making behaviors, such as impulsive decision-making.

Impact statement

Existing research on tourist decision-making predominantly focuses on the pre-decision, decision-in-progress, and post-decision stages. This study shifts attention to the pivotal, yet often overlooked transitional moments between these phases. Two novel concepts, the Decision-Making Threshold (DMT) for the start point of decision and the Information Loop Limit (ILL) for the endpoint, demonstrate that individuals differ in their tendencies to initiate and conclude the decision-making process. By focusing on the start instance, marketing strategies can be tailored to effectively steer potential customers into decision-related information processing, while for the ending moment, interrupting or extending decision-making process can be realized. The new constructs can be included as viable dimensions in the development of user profiles

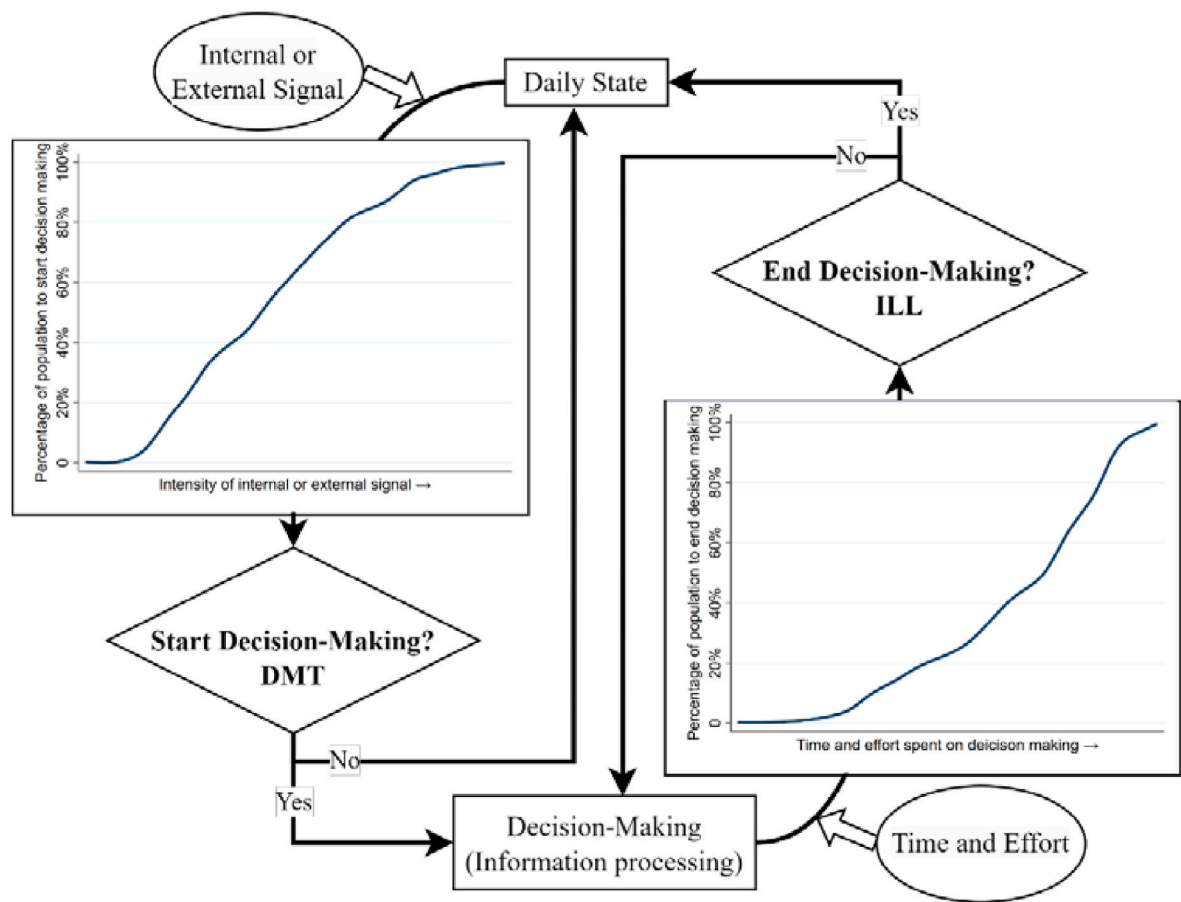


Fig. 4. DMT and ILL that control the start and end of decision-making¹.

and the segmentation of decision-makers, including rational frequent travelers, travel career ladder climbers, less-experienced low-information decision-makers, and cautious senior decision-makers. Practitioners can develop targeted strategies to provide appropriate types, content, and volumes of information to suit each target segment.

CRediT authorship contribution statement

Ruizhe Fang: Writing – original draft, Visualization, Methodology,

Formal analysis, Data curation, Conceptualization. Li Pan: Writing – review & editing, Validation, Methodology, Funding acquisition.

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Appendix A

Appendix A1DMT item-eliminating process

Item	Item wording	Mean	SD	Scale 1		Scale 2			Scale 3		
				ITC	Cronbach's α if item deleted	ITC	Cronbach's α if item deleted	AIC	ITC	Cronbach's α if item deleted	AIC
1	When I see travel-related information, I become aware of my travel needs and initiate the process of making travel consumption decisions. (R)	2.229	0.961	0.314	0.709						
2	When I recall my previous travel experiences, I initiate the decision-making process for the next travel consumption. (R)	2.495	1.040	0.328	0.707	0.024	0.742	0.029			
3	Whenever I have free time, I start planning for the next trip. (R)	2.271	1.037	0.299	0.710	0.034	0.740	0.034			
4	When I want to participate in a travel activity, I immediately start making travel consumption decisions. (R)	2.346	0.953	0.227	0.717						

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Item	Item wording	Mean	SD	Scale 1		Scale 2			Scale 3		
				ITC	Cronbach's α if item deleted	ITC	Cronbach's α if item deleted	AIC	ITC	Cronbach's α if item deleted	AIC
5	When I hear about interesting places, I start the process of making travel consumption decisions. (R)	2.304	0.970	0.426	0.698						
6	Whenever I feel like traveling, I initiate the process of making travel consumption decisions. (R)	2.206	0.909	0.337	0.707						
7	When I see travel-related advertisements, I start contemplating travel consumption decisions. (R)	2.444	1.065	0.310	0.709	-0.049	0.754	-0.011			
8	The joy brought by travel prompts me to initiate the process of making travel consumption decisions. (R)	2.210	0.936	0.299	0.710						
9	When I come across travel-related recommendations, I consider specific travel plans that I can implement. (R)	2.327	0.998	0.340	0.706						
10	I do not initiate the process of making travel consumption decisions simply because I have the idea of traveling.	3.360	1.214	0.412	0.697	0.629	0.633	0.315	0.686	0.848	0.543
11	When I see travel-related advertisements, I do not start making travel consumption decisions.	3.304	1.126	0.385	0.700	0.589	0.645	0.297	0.635	0.857	0.508
12	Recalling the pleasure brought by previous travel experiences does not make me start planning future trips.	3.346	1.323	0.266	0.716	0.510	0.657	0.258	0.625	0.860	0.500
13	When I realize that I have a desire to travel, I do not initiate the process of making travel consumption decisions.	3.201	1.204	0.253	0.716	0.513	0.658	0.255	0.664	0.852	0.526
14	When I see information regarding travel destinations, I do not start making travel consumption decisions.	3.220	1.181	0.377	0.701	0.632	0.634	0.313	0.719	0.842	0.565
15	Even if friends discuss travel topics, I do not start making travel consumption decisions.	3.173	1.220	0.314	0.709	0.584	0.643	0.289	0.714	0.843	0.560
	Cronbach's α			0.729		0.689			0.873		

Note: All item scores range from 1 to 5. The reverse-scored items are marked with (R). Statistics that caused the deletion of items are illustrated in bold. ITC = item-total correlation. AIC = Average inter-item correlation.

Appendix A2ILL item-eliminating process

Item	Item wording	Mean	SD	Scale 1		Scale 2			Scale 3		
				ITC	Cronbach's α if item deleted	ITC	Cronbach's α if item deleted	AIC	ITC	Cronbach's α if item deleted	AIC
1	Before making travel consumption decisions, I spend a considerable amount of time and effort in information processing.	3.771	0.959	0.275	0.824						
2	Regarding travel consumption, I dedicate a great deal of time and effort to search for and compare various decision alternatives.	3.813	0.980	0.304	0.823						
3	During the process of making travel consumption decisions, I engage in extensive mental labor.	3.612	0.961	0.135	0.831						
4	I continuously focus on processing relevant information related to travel consumption decisions until a decision must be made.	3.720	0.967	0.131	0.831						
5	In the travel decision-making process, I invest more time and effort in information processing than most people do.	3.551	0.995	0.176	0.829						
6	I usually spend a significant amount of time and effort comparing different travel products/services before making a decision.	3.654	0.970	0.179	0.829						
7	My travel consumption decisions are based on a substantial amount of relevant information.	3.841	0.961	0.311	0.822						

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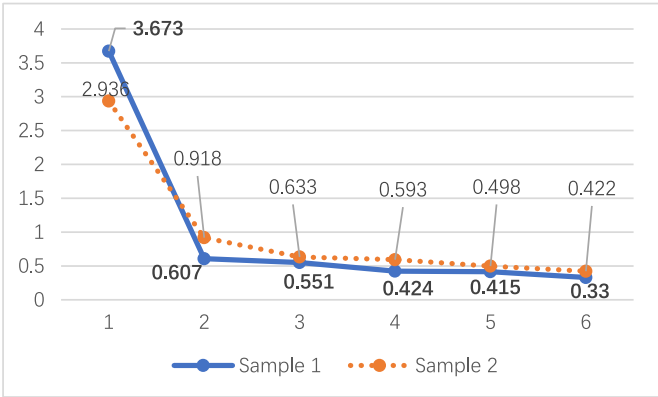
¹ The curves are derived from the response scores of all respondents in the three samples, primarily to demonstrate the differences in DMT and ILL on individuals. With increasing intensity, internal or external signal can break through the DMT for more individuals, who initiate their decision-making process. As the decision-making process progresses with augmented time and effort, more individuals reach their ILL, ending their decision-making activities.

(continued)

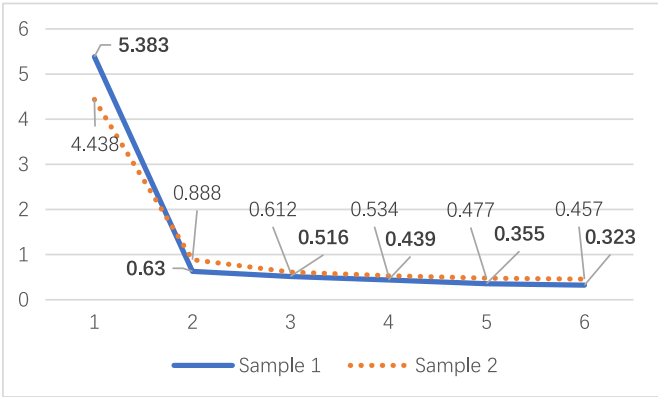
Item	Item wording	Mean	SD	Scale 1		Scale 2			Scale 3		
				ITC	Cronbach's α if item deleted	ITC	Cronbach's α if item deleted	AIC	ITC	Cronbach's α if item deleted	AIC
8	I consider more information before making travel decisions than most people do.	3.818	0.939	0.290	0.823						
9	I am unwilling to spend time and effort in information processing for travel consumption decisions. (R)	2.897	1.270	0.634	0.802	0.756	0.917	0.686	0.760	0.919	0.625
10	Making travel-related decisions does not require a great deal of information. (R)	2.832	1.198	0.501	0.811	0.672	0.923	0.618	0.670	0.926	0.558
11	I feel that spending too much time processing information pertaining to travel decisions is unnecessary. (R)	2.893	1.215	0.637	0.802	0.785	0.915	0.710	0.795	0.917	0.651
12	I do not have much time or energy to dedicate to information processing for travel consumption decisions. (R)	2.668	1.178	0.627	0.803	0.797	0.915	0.725	0.800	0.916	0.659
13	I do not let travel consumption decisions consume my time or energy. (R)	2.911	1.320	0.613	0.803	0.757	0.917	0.686	0.768	0.919	0.630
14	During the decision-making process of travel consumption, I tend to be distracted by other thoughts. (R)	2.692	1.087	0.360	0.820	0.552	0.930	0.520			
15	Travel consumption decisions involve too much information for me to consider. (R)	2.682	1.188	0.601	0.805	0.722	0.919	0.661	0.712	0.923	0.591
16	I only focus on a small amount of information during the travel consumption decision-making process. (R)	2.902	1.291	0.558	0.807	0.740	0.918	0.674	0.735	0.921	0.606
17	I cannot devote too much time and effort to information processing pertaining to travel consumption decisions. (R)	2.640	1.082	0.663	0.802	0.836	0.913	0.755	0.825	0.915	0.675
	Cronbach's α			0.814		0.927			0.930		

Note: All item scores range from 1 to 5. The reverse-scored items are marked with (R). Statistics that caused the deletion of items are illustrated in bold. ITC = item-total correlation. AIC = Average inter-item correlation.

Appendix B



Appendix B1Scree plot from two samples for DMT scale



Appendix B2Scree plot from two samples for ILL scale

Appendix. CSupplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.tourman.2024.104918>.

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